

A SPECIAL EDITION OF

MERIDIAN

Sim racing — an ultra deep dive

 A racing-simulator rig with wheel, pedals and a wraparound display

Source: Wikimedia Commons — a racing simulator rig at a public gaming expo, representative of the home-cockpit class this issue covers, not a specific reader's build.

THE BAKE-OFF

Seven platforms, ranked with an asterisk — the real hierarchy, not the marketing one

THE CONSOLIDATION

One company now owns the wheel, the pedals and the seat underneath them

THE LONG READ

A CT Hub studio, two national titles, and Singapore's actual sim-racing scene

THIS SPECIAL

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FROM THE DESK

"If everything seems under control, you're not going fast enough."

— Mario Andretti, racing driver¹

Andretti spent a career at the edge of grip in cars costing what a house does. The line still governs a hobby that now lets anyone find that same edge for the price of a game console — which is exactly why this issue exists: the reader tapped this morning's Rabbit Hole page and asked for everything the three-page version had no room for.

SIM RACING — AT A GLANCE

METRIC	FIGURE
Genre origin	Papyrus, 1989
Dominant subscription sim	iRacing, since 2008
iRacing annual fee, from 18 Aug 2026	US\$124.50
Cheapest bundled DD wheelbase	~US\$299–379
2026 torque price-floor break	Moza R21/R25 Ultra, \$699/\$899
Aug 2026 hardware consolidation	Corsair buys Trak Racer, ~\$44–50M
Singapore's leading org	Legion of Racers, CT Hub

Figures as researched for this edition, Aug 2026; see the sections named for full sourcing.²

WHY THIS SPECIAL EXISTS

No. 66's Rabbit Hole gave sim racing three pages this morning — the home builders, the Corsair–Trak Racer deal, the price-floor break, a one-line sceptic's jab. A reader tapped that page's feedback row within hours and asked for the rest. This special assumes that coverage as read and goes past it: the full history, the complete gear ladder, the racing and league side, the community and its fights, a harder run at the sceptic's case, and Singapore's actual scene, not a mention of it.³

THE RABBIT HOLE, CONTINUED

The Invitation

Three pages this morning covered the home builders, a \$50 million acquisition and a price-floor break. This is the other twenty-one.

Sim racing is the only hobby in this ledger where the top of the market and the bottom of the market run the exact same software, on the exact same virtual track, at the exact same time. A teenager on a \$299 wheel and a factory GT3 driver on a \$6,000 rig can be gridded next to each other in the same iRacing race tonight, and the sim does not know the difference — only the stopwatch does.¹ That single fact is the whole appeal and the whole argument, and it runs through every section of this book.

No. 66's Rabbit Hole gave the hobby three pages this morning: the men who build an entire car's worth of hardware in a spare room, Corsair's acquisition of Trak Racer five days ago, the Moza R21 and R25 Ultra breaking a torque-per-dollar floor, and a closing jab that a \$2,000 wheel doesn't lap a mid-pack racer any faster than a \$500 one.² All of that stands — this book assumes it as read rather than repeating it, and spends its twenty-four pages on what three pages had no room for: where the genre actually came from, what a direct-drive motor is doing that a belt-driven wheel cannot, an honest ranking of the seven platforms serious racers actually run, the complete gear ladder from bundle to boutique, the racing and esports side with its real crossover drivers, the community map and the fights inside it, a harder cross-examination of that closing sceptic's line, and — because it genuinely exists — Singapore's own competitive scene.

The hobby's economics changed meaningfully in the same week this book was commissioned. On 5

August, Corsair Gaming — which already owned Fanatec, the hardware line most serious racers grew up buying — announced it was acquiring Trak Racer, a Melbourne-founded maker of cockpits, seats and motion rigs, for cash consideration Corsair's own SEC filing puts at roughly US\$43.7 million net of acquired cash, against a trade-press headline figure of "\$50 million" that appears to fold in a further \$10 million earn-out.³ Two days earlier, Moza Racing — one of the last large independents — had already broken a price floor of its own: a 21Nm direct-drive wheelbase at \$699 and a 25Nm one at \$899, undercutting the previous benchmark at that torque class by roughly half.⁴ Both stories get the full treatment on p15.

"Buy-once-cry-once" is the phrase this magazine's Rabbit Hole desk reaches for whenever a hobby has a real spending ladder rather than a fixed price tag — home espresso, mechanical keyboards, hi-fi. Sim racing earns the label as cleanly as any hobby this book has covered: the same core activity, lapping a virtual circuit, is available at \$299 and at \$50,000, with every rung between the two genuinely improving something real (feel, precision, immersion) rather than just signalling spend. What makes it different from most buy-once-cry-once hobbies is the software underneath never changes with the hardware — a \$299 setup and a \$50,000 one run the identical physics engine, which is precisely the tension The Sceptic, on p23, spends two pages taking seriously rather than dismissing.

THE COMMUNITY READ

"If you're slower than about 105% of the fast guys' pace, the wheel isn't your limitation — racecraft and pedal work are," runs one widely echoed line from hardware-comparison content weighing a cheap wheel against a premium direct-drive base — a claim this book tests properly rather than just repeating, on p23.⁵

— paraphrased across sim-racing hardware review content, 2026

300,000

SIMULTANEOUS ACTIVE IRACING ACCOUNTS, CLAIMED PEAK — SINGLE-SOURCE, UNCONFIRMED

SOURCES 1 iRacing.com, license-class and matchmaking structure, via iRacing Support documentation. 2 MERIDIAN No. 66, The Rabbit Hole, 8 Aug 2026. 3 Corsair Gaming Inc., Form 10-Q, period ended 30 Jun 2026, subsequent events note, via SEC EDGAR; Traxion.gg, "Trak Racer to operate as a Fanatec sub-brand in \$50m Corsair deal," 5 Aug 2026. 4 Traxion.gg, "MOZA R21 Ultra DD Wheel Base review: Affordable torque," Aug 2026; Bsimracing, "Moza Racing R21 Ultra and R25 Ultra: True Torque Wheelbases," Aug 2026. 5 DG Edge, "Does Expensive Sim Racing Wheels Make You Faster?," undated, 2026 context.

What makes this a genuine hobby rather than just a game category is the same thing that makes model railways or serious hi-fi genuine hobbies: a buy-once-cry-once economics that never actually stops, a craft dimension (car setup, pedal technique, cockpit-building) that rewards obsession, a real competitive ladder with its own culture and politics, and a community that argues about all of it constantly, in public, with receipts.¹ Every one of those threads gets its own room in this book.

A word on scope. This is not a buying guide dressed up as journalism — every price and spec named in the sections that follow is sourced and dated, and where a figure could not be independently confirmed this issue says so plainly rather than printing a guess. Sim racing hardware pricing moves fast and manufacturer sites were unreachable from this session's own network; every figure below traces to a specific outlet, and single-source numbers are flagged as such rather than presented with false confidence.²

The shape of the book: The Circuit and The Machine (p5–p8) cover where the genre came from and what the technology is actually doing. The Bake-Off (p9) ranks the seven platforms a serious racer would actually choose between today. The Rig and The Build (p11–p14) are the complete gear ladder — wheelbases, pedals, cockpits, motion, displays, and the custom-maker tier. The Consolidation (p15) is the industry and economics story, including what it actually costs to import this hardware into Singapore. The League (p17) is the racing and esports side, with the real drivers who turned a controller into a contract. The Community (p19) maps the forums, Discords, creators and venues, honestly —

including the ones this book could not verify. The Wars (p21) is the culture's real fights: pay-to-race, toxicity, cheating scandals, and whether any of this counts as sport. The Sceptic (p23) is the hardest test of the hobby's own mythology. And the Long Read (p25) is Singapore's actual competitive scene — a studio at CT Hub with two national esports titles to its name, mined the way this magazine's Undercurrent mines any live regional story.

A note on what this book is not. It is not a review of any single wheelbase or pedal set — no unit was tested for this issue, and every performance claim about specific hardware traces to a named external review, not this desk's own hands-on time. It is also not a defence of the hobby against the obvious criticism that it is an expensive way to sit in a chair pretending to drive — that criticism gets its fair hearing on p23, argued honestly rather than waved away. What this book tries to be instead is the thing a genuinely curious newcomer with a few hundred dollars and a free weekend would actually want: an honest map of where the money goes, what it buys, who else is doing this, and whether any of it is worth doing at all.

One structural choice worth flagging up front: this book runs no product recommendations phrased as "buy this." Every gear section instead states what a given price tier reliably delivers, sourced to independent reviews and listings, and leaves the decision where it belongs — with the reader's own budget and how seriously they expect to take the hobby. A magazine that reviews nothing it hasn't tested has no business telling anyone what to buy; it can only tell them, honestly, what the market says a given price actually gets.

27pp

ELEVEN SECTIONS, ONE TOPIC, NO DAILY DESKS

THE CROSS-EXAMINATION, EARLY

Not every claim in sim racing culture survives contact with evidence. This book flags every single-source figure, every unverifiable claim and every piece of vendor-adjacent "research" as such, rather than letting a hobby's own enthusiasm launder its weaker arguments into fact. Where the record is thin — a controversial thread, a named creator, a venue's opening hours — this book says so instead of filling the gap.³

SOURCES 1 Community-map and culture research compiled for this edition, Aug 2026 — see p19–p22. 2 Hardware research compiled for this edition, Aug 2026 — see p11–p14; note manufacturer sites (Fanatec, Simucube, Moza, Simagic) were unreachable this session, so figures are cross-checked across retailer and press listings rather than primary product pages. 3 Editorial standard, this edition.

HISTORY

The Circuit

A 1987 educational-software company invented the genre by accident. Its bankruptcy is the reason the dominant platform today exists at all.

Sim racing's family tree starts at Papyrus Design Group, founded in Watertown, Massachusetts in 1987 by David Kaemmer and Omar Khudari — a studio whose first credited work together was educational Apple II software, not games.¹ Papyrus's 1989 title *Indianapolis 500: The Simulation*, published by Electronic Arts, is widely cited as the point racing software crossed from arcade game into simulation proper: car setup changed how the vehicle handled, and there were no arcade shortcuts.² A licensing deal with NASCAR in 1994 sold over a million copies and made the studio's name.³

Papyrus's 1998 release, *Grand Prix Legends* — a recreation of the 1967 Formula One season — is still argued over by sim racers today for a physics model reviewers and enthusiasts alike describe as never quite surpassed: cars could genuinely drift and unsettle over kerbs, there were no driver aids of any kind, and the fourteen-mile, 174-corner original Nürburgring Nordschleife was in the box.⁴ Its 2003 successor, *NASCAR Racing 2003 Season* — refined, per one retrospective, "under consultation with NASCAR team engineers" — was Papyrus's final release. Both titles retain active modding communities today, more than two decades on, still adding cars and tracks to software last formally supported in 2004.⁵

Papyrus closed in May 2004 after Electronic Arts acquired exclusive NASCAR game rights. David Kaemmer bought the source code and assets himself and, with businessman John Henry — principal

owner of Fenway Sports Group and co-owner of the NASCAR team Roush Fenway Racing — built a new company on that foundation. It launched publicly on 28 August 2008, under a name that is now the closest thing the hobby has to a default: iRacing.⁶

What made Papyrus's engineering different from arcade racers of the same era was never really the graphics — it was the refusal to shortcut the driving model even when the hardware of the day made that refusal expensive to render. *Indianapolis 500: The Simulation* modelled setup changes (gear ratios, wing angles, tyre stagger) that visibly altered how the car behaved on track, at a time when most competing titles treated a car as a single fixed handling profile with a paint job swapped on top.⁸ That single design decision — physics first, presentation second — is the thread every studio profiled on p6 inherited, whichever branch of the family tree they descend from.

Sources genuinely disagree on iRacing's own founding year — flagged properly in the cross-examination below — but there is no disagreement on the continuity of the code itself. iRacing's oval-racing physics, per multiple retrospectives, trace in a direct engineering line back through *NASCAR Racing 2003 Season* to the original 1994 *NASCAR Racing* engine — meaning a driver turning laps in iRacing's Truck Series tonight is, at some level, still running physics code with a genealogy stretching back thirty-two years, longer than most of this hobby's current racers have been alive.⁹

1987

YEAR PAPYRUS DESIGN GROUP WAS FOUNDED IN WATERTOWN, MASSACHUSETTS

TWO TITLES, STILL RACED

Grand Prix Legends and *NASCAR Racing 2003 Season* are both playable on Windows 11 today, entirely through community effort — no publisher has touched either in over twenty years. OverTake.gg's own retrospective calls NR2003 "Papyrus' Masterpiece."⁷

SOURCES 1 Wikipedia / GameSpot, "History of Papyrus Racing Games," via research compiled Aug 2026. 2 Wikipedia, dosdays.co.uk, myabandonware, on *Indianapolis 500: The Simulation* (1989). 3 GameSpot, "History of Papyrus Racing Games." 4 Fanatec Explorer blog, "Evolution of Sim Racing" series; RaceSimCentral, on *Grand Prix Legends* (1998). 5 OverTake.gg, "NASCAR Racing 2003: Papyrus' Masterpiece?," retrospective; RaceSimCentral. 6 OverTake.gg, "15 Years Already?" iRacing retrospective; Automobile Fandom wiki. 7 OverTake.gg retrospective, as above. 8 Wikipedia, dosdays.co.uk, as source 2. 9 OverTake.gg iRacing retrospective; GameSpot "History of Papyrus Racing Games," as sources 3 and 6.

iRacing was not the only branch. Image Space Incorporated, founded in Ann Arbor, Michigan in 1992 by Kurt Kleinsorge with partners Joseph Campana and Gjon Camaj, built its own physics engine — gMotor2 — and released rFactor in August 2005: an early download-distributed, deeply moddable sim shipped with fictional cars and tracks by default, designed to be rebuilt by its own community.¹ That studio's engine and its Brazilian licensee, Reiza Studios — begun as an rFactor modding team led by Renato Simioni — eventually produced Automobilista and, on a different engine entirely, Automobilista 2.²

A separate lineage runs through Kunos Simulazioni, founded in 2005 near Rome by Stefano Casillo and Marco Massarutto, whose Assetto Corsa engine became the basis for Assetto Corsa Competizione — the officially licensed GT World Challenge Europe sim — and, in 2025, the early-access Assetto Corsa EVO. Casillo left Kunos in 2020 and rejoined as CTO in March 2025, reuniting the studio's two founders.³ A fourth lineage, Ian Bell's Simbin Studios and then Slightly Mad Studios, produced Project CARS before Bell sold the studio to Codemasters for a reported US\$30 million in November 2019; its "Madness Engine" was later licensed by Reiza for Automobilista 2.⁴

Console racing followed its own path. Kazunori Yamauchi began Gran Turismo inside Sony's Japan studio with a five-person team that grew to seventeen before being restructured into Polyphony Digital in April 1998; the first game shipped in December 1997 after five years' development and

became the best-selling PlayStation title of its era.⁵ Microsoft's Forza line runs in parallel on Xbox and PC. Both console franchises trade some simulation depth for accessibility — a tension explored fully on p9.

What ties all four lineages together is more interesting than any single studio's story: every one of them began as a small, obsessive team chasing physics fidelity most of the games industry considered commercially irrelevant, and every one of them eventually got bought, restructured or absorbed by something larger once the obsession proved to have a real market. Papyrus sold to Sierra in 1995. Slightly Mad sold to Codemasters in 2019. Kunos sold to Digital Bros in 2017 for a reported €4.3 million. Studio 397 — the modern custodian of the ISI/rFactor lineage — was bought into Motorsport Games in 2021, a company now disclosing an \$86.3 million deficit, discussed fully on p9. Consolidation is not new to this hobby's software side; The Consolidation, p15, is simply the moment it caught up with the hardware side too.⁷

One studio deliberately left off this family tree deserves a line of its own: Papyrus's own DNA also runs, more distantly, through the arcade-leaning end of the genre via staff and licensing relationships this book's research could not fully trace — a reminder that a forty-year-old genre's exact lineage is genuinely contested in places, not a tidy chart. Where this book could not confirm a claimed connection, it left the connection out rather than drawing a confident-looking line between two boxes.

1998

GRAND PRIX LEGENDS' PHYSICS STILL SETS THE BAR ENTHUSIASTS ARGUE OVER

IN BRIEF — THE FOUR LINEAGES

Papyrus → **iRacing** Kaemmer/Khudari, 1987; code bought back after 2004 closure.

ISI → **rFactor / Reiza** Kleinsorge, 1992; gMotor2 engine, moddable by design.

Kunos Simulazioni Casillo/Massarutto, 2005; Assetto Corsa and ACC.

Simbin → **Slightly Mad** Ian Bell; Project CARS, later Automobilista 2's engine.

THE CROSS-EXAMINATION

Sources disagree on iRacing's own founding year — some place the company's founding in 2003, before Papyrus's 2004 closure, others in 2004 itself. The most likely reading is that 2003 marks initial planning or registration and 2004 the point Kaemmer began building on the reacquired Papyrus code — but this book could not resolve the discrepancy to a single date, and prints both rather than picking one to sound authoritative.⁶

SOURCES 1 Wikipedia, "Image Space Incorporated"; OverTake.gg, RaceSimCentral. 2 Automobilista 2 Fandom wiki; BSImRacing. 3 Wikipedia, "Kunos Simulazioni"; OverTake.gg/Traxion.gg coverage of Casillo's 2025 return; BoxThisLap. 4 Wikipedia, "Slightly Mad Studios"; Traxion.gg, "The rise and fall of Project CARS"; GTPlanet; VGC. 5 Wikipedia, "Gran Turismo (series)"; PlayStation Blog, "Origin Story." 6 iracing.com "History" page and Automobile Fandom wiki, both via research compiled Aug 2026 — dates as printed, unresolved between sources. 7 Wikipedia, "Slightly Mad Studios"; Wikipedia, "Kunos Simulazioni"; Insider Gaming, OverTake.gg on Motorsport Games' finances, as sourced on p9.

TECHNOLOGY

The Machine

A servo motor from industrial robotics, a tyre model borrowed from a 1980s Dutch engineer, and a physics clock that runs six times faster than the screen.

"Force feedback" is the wheel pushing back — simulating what a real steering rack transmits from tyre to driver's hands. Cheap wheels do this with gears (a physically notchy feel, roughly 15–25ms of latency) or belts and pulleys (smoother but with slight flex-induced softness, roughly 10–15ms). Direct-drive wheels remove the gears and belts entirely and connect the wheel shaft straight to a brushless servo motor — the same class of motor used in industrial robotics and CNC machinery — giving what the hardware press consistently calls "true 1:1" feedback, with latency cited around 3–8ms, widely described as imperceptible.¹ The buyer's ladder for these motors runs the whole of p11.

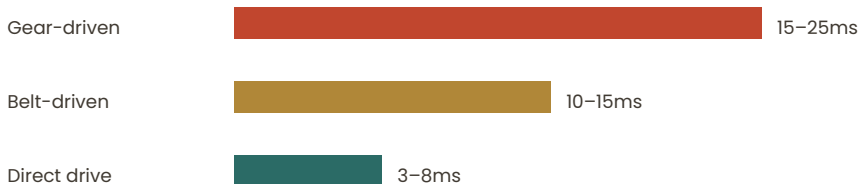
What separates a "sim" from an arcade racer is mostly the tyre model. Sim-grade titles simulate slip angle — the gap between where a tyre points and where it is actually travelling — plus load transfer, camber and temperature, typically through variants of the Pacejka "Magic Formula," a model built to

capture the nonlinear handoff from static grip to sliding friction across a range of loads.² Automobilista 2 layers this further with a three-part SETA model — carcass, tread contact patch, and heat transfer, simulated separately.³ Gran Turismo 7's well-regarded but shallower model — no engine stalling, no clutch bite point, limited tyre-pressure simulation, fully automated pit stops — is why GTPlanet's own community routinely files it as "simcade" rather than sim, discussed fully on p9.⁴

iRacing's physics engine runs at a fixed 60Hz — every physics frame gets 16.7 milliseconds to compute. A 240Hz monitor does not make the underlying physics more accurate; it only smooths how the last 60Hz state is displayed. Force feedback itself is also 60Hz natively, though a community tool and native support on some direct-drive hardware can reconstruct the signal up to 360Hz — a genuine and often-argued technical point about how "raw" the feel is on premium hardware versus stock.⁵

FORCE-FEEDBACK LATENCY BY DRIVE TYPE

Widely-cited enthusiast-press figures, milliseconds — illustrative, not lab-measured



Source: MOZA Racing, Logitech G, Fanatec Explorer, buyer's-guide press aggregation, 2026 — see sources, this page.

Fig. 1 — Cited force-feedback latency by wheel drive type. Original figure, MERIDIAN, built from sourced ranges.⁶

THE COMMUNITY READ

"Direct drive ruined me for anything else" is the single most repeated line in force-feedback discussion threads — not a claim about lap time, enthusiasts are usually careful to say, but about how much more information the wheel communicates before a slide actually happens.⁷

SOURCES 1 MOZA Racing blog; Logitech G "Discover" page; Fanatec Explorer blog; buyer's-guide aggregation, research compiled Aug 2026. 2 Tyre-physics explainer sources aggregated for this edition. 3 Automobilista 2 Fandom wiki, on the SETA tyre model. 4 GTPlanet forum discussion; BoxThisLap analysis, on GT7's "simcade" reputation. 5 pingpackettest.com; simracingcockpit.gg technical deep-dive; Granite Devices/Simucube community forum; Overtake.gg thread on iRacing physics tick rate. 6 As sources 1, this page; figures are enthusiast-press aggregates, not independently lab-measured — printed as illustrative, not precision engineering data. 7 Community-forum sentiment aggregated for this edition, characterised rather than tied to one named post.

Pedals split the same way wheels do, on a different axis: travel versus force. A potentiometer pedal measures how far the pedal physically moves; a load-cell pedal measures the pressure applied to the pedal face, regardless of travel — which is closer to how a real car's hydraulic brake actually works, where a firm car barely moves the pedal at all under heavy braking.¹ The mechanism explanation is consistent across every source this book checked; the specific claim that load cells buy "one to two seconds a lap" for an intermediate driver is not — it traces to no named study and should be read as a widely repeated but unverified figure, not a measured result.²

One genuine study did surface in this book's research: a paper on comparative load-cell and Hall-effect (potentiometer-style) brake-sensor performance, indexed on PMC and apparently published in *PLoS ONE* in January 2024, describing a repeated-measures test of twenty experienced sim racers. Both search summaries used to source this book agree the study found faster lap times and more "faster-driver-like" braking behaviour — greater brake-force amplitude, longer trail-braking — on the load-cell sensor.³ This is flagged rather than treated as settled: this book could not open the primary text directly to confirm authorship, and it is the single piece of evidence in this whole issue that reads as genuine peer-reviewed research rather than hardware-press consensus — which is itself worth noting about how thin the hobby's actual evidence base is, a theme p23 returns to directly.

Telemetry — throttle, brake and steering traces, tyre temperature and wear per corner, suspension travel, lap and sector splits — is the same discipline real motorsport engineers use, applied by hobbyists to

their own driving. Named tools include MoTeC, the same brand real teams use, and Virtual Racing School's coaching platform; the purpose, per multiple sim-coaching sources, is explicitly "the same type of engineering feedback that professional drivers rely on," turned inward on an amateur's own laps.⁴

The practical effect of telemetry culture is a hobby that argues with itself constantly using its own data rather than opinion — a driver claiming a car is "loose off the throttle" can be shown, in the same session's overlay, whether the rear actually stepped out earlier than a faster reference lap, or whether the perception was simply a scarier-feeling but equally fast line. That habit of checking feeling against trace data is, per multiple coaching sources, the single fastest way a newcomer actually improves — faster, by a wide margin, than any hardware purchase this book prices on p11-p14. It is also, not coincidentally, free: every platform on p9 ships some form of built-in telemetry overlay.

One further distinction worth holding onto through the rest of this book: "sim" and "realistic" are not quite synonyms. A tyre model can be mathematically sophisticated (sim-grade, per the definition on p7) while still being tuned by its developer for a specific feel — grippier, more forgiving, more punishing — that has nothing to do with fidelity and everything to do with house style. iRacing, ACC and rFactor 2 each have a recognisably different "feel" to experienced racers despite all three qualifying as genuinely sim-grade by the tyre-model test — a distinction The Bake-Off, next, treats as a feature to choose between, not a flaw to resolve.

GLOSSARY — IN BRIEF

FFB Force feedback — the wheel's resistance/kick, simulating tyre grip through the steering.

DD Direct drive — motor connected straight to the wheel shaft, no belts or gears.

Load cell A pedal sensor measuring applied force rather than travel distance.

iRating / SR iRacing's skill rating and Safety (clean-driving) rating — see p17.

The wheel updates every 16.7 milliseconds. Your eyes are the slow part.

THE CROSS-EXAMINATION

"One to two seconds a lap" from a load-cell upgrade is repeated constantly and traces to no named study — the one real research finding this book located supports a real effect on braking consistency, not a specific lap-time number, and this book prints the honest version rather than the round one.⁵

SOURCES 1 Asetek SimSports, "What Are Load Cell Pedals?"; OverTake.gg forum thread; Racemarket.net blog; FATBLACKCAT.io, research compiled Aug 2026. 2 As source 1 — the specific lap-time figure traces to no named study across sources checked. 3 PMC (PMC12251626) / DOAJ index entry consistent with a *PLoS ONE*, Jan 2024 paper on load-cell vs. Hall-effect brake sensors — primary text unreachable this session; treat as credible-but-unconfirmed pending direct verification. 4 MoTeC; Virtual Racing School; sim-coaching site aggregation, research compiled Aug 2026. 5 As sources 2–3, this page.

THE PLATFORMS

The Bake-Off

A subscription oval-and-road giant, a financially fragile endurance sim critics love, and a console flagship the hardcore call "simcade." No single winner — a fit exercise.

There is no single "best" sim racing platform, because the seven that matter are not really competing for the same racer. What follows ranks each on what it is actually for, not against a single scorecard — the asterisk is the point.

Ordering seven products against each other invites the obvious challenge: ranked by what? This list is ordered by a single question — which platform would a genuinely serious newcomer, with no existing

loyalty, most likely end up on within a year — not by concurrent-player count, review score, or physics purity taken alone. iRacing tops it for structural reasons (the ladder, the money, the official world championships) that have nothing to do with which sim feels best on the wheel; several racers this book's research surfaced rate ACC or Le Mans Ultimate's actual driving feel higher while still spending most of their competitive hours on iRacing, because the ladder is where the racing that matters happens.

- 1. iRacing — the ladder.** Pure subscription, no free base game: \$124.50/year from 18 Aug 2026 (up from \$110, the second price rise since 2008), plus \$11.95 per car and \$11.95–\$14.95 per track on top.¹ Full detail and the "pay to race" argument on p15 and p21. Unmatched for its license ladder and official world championships (p17) — the only platform here with a genuine sporting pyramid.
- 2. Assetto Corsa Competizione — the GT reference.** Kunos's officially licensed GT World Challenge Europe sim, full release May 2019, one-time purchase. Peaked at 12,376 concurrent Steam players (1 Apr 2024, likely event-driven); recent 30-day trend down roughly 42% from that peak, current live counts in the low thousands.²
- 3. Le Mans Ultimate — the best physics, the shakiest publisher.** Studio 397's WEC/Le Mans-licensed sim, built on the rFactor 2 engine, reached full 1.0 release 22 July 2025 after eighteen months in early access. Publisher Motorsport Games reported an \$86.3 million deficit as of 30 Sept 2024 and a market cap near \$8.25 million — genuinely uncertain long-term footing behind a title the hardcore rate highly.³
- 4. rFactor 2 — ageing, still respected.** Studio 397's older platform, acquired into Motorsport Games in 2021. Concurrent Steam players down roughly 92% from a November 2023 peak of 1,586, to an estimated 134–349 — but veteran racers still cite its physics and force feedback as genuinely strong.⁴
- 5. Gran Turismo Sport — the on-ramp.** Console-exclusive (PS4/PS5), March 2022, one-time purchase plus microtransactions. GTPlanet's own community routinely tags it "simcade" — no engine stalling, no clutch bite point, automated pit stops — even as some real-world drivers are quoted saying the base handling "feels shockingly right."⁵
- 6. Forza Motorsport (2023) — the reboot that split its own audience.** Critical scores solid (Metacritic 83–84), but Steam user reviews only 43% positive and peak concurrent players under 5,000 at launch — reported as Microsoft's weakest 2023 gaming release by that metric, largely over a car-progression grind system.⁶
- 7. Automobilista 2 — the value pick.** Reiza's Madness-Engine title with a layered SETA tyre model; frequently discounted (seen as low as \$3.99 in a 90%-off sale) though its standing list price could not be confirmed this session.⁷

Seven platforms. No winner. Only a better or worse fit.

THE CROSS-EXAMINATION

Assetto Corsa EVO — Kunos's early-access successor, launched Jan 2025 with Stefano Casillo back as CTO — is deliberately left off this ranking. Its 1.0 release was targeted for Fall 2025, then "not before Summer 2026" per publisher Digital Bros's own financial reporting, and no confirmed full-release date had surfaced in this book's research as of 8 August 2026. Ranking an unfinished product against seven shipped ones would flatter it; it is noted here, honestly, as still cooking.⁸

SOURCES 1 BoxThisLap, "iRacing Subscription Price Increase, August 2026"; OverTake.gg; Traxion.gg. 2 SteamDB; Steambase; BoxThisLap, "Sim Racing on Steam." 3 OverTake.gg; Insider Gaming; Motorsport Games SEC filing (EX-99.1) via StockTitan; Motorsport Week; FIA WEC. 4 SteamDB; Tracker.gg; opinion coverage aggregated, treated as colour not fact. 5 GTPlanet forum discussion; BoxThisLap analysis. 6 Wikipedia, "Forza Motorsport (2023 video game)"; GameWorldObserver; GTPlanet review. 7 Steam listing, Summer Sale 2025 price observed; base price unconfirmed this session. 8 GTPlanet; OverTake.gg; gamespress.com; 505games.com — full-release status as of 8 Aug 2026 unconfirmed, phrased as such rather than asserted either way.

The practical upshot for a newcomer: pick the platform by what you want to do, not by review score. Want a structured ladder with real stakes and official world championships to climb toward — accept the subscription-plus-content economics and start on iRacing. Want the single best-regarded GT experience with a real license and no ongoing content bill — ACC. Want the deepest current physics and don't mind betting on a financially fragile publisher — Le Mans Ultimate. Want to play on a console you already own, casually, without buying a wheel at all — Gran Turismo 7 or Forza. Most serious racers this book's research surfaced end up owning two or three of these rather than one — a road/GT platform for structured racing, a console or arcade-adjacent title for casual laps, and increasingly a WEC-flavoured title for the endurance itch Le Mans Ultimate and ACC both scratch differently.¹

One structural point threads all seven: none of them are free, and the "free-to-play" framing occasionally applied to sim racing online is a category error inherited from other genres. Even iRacing's own most-populated series, the Mazda MX-5 Cup, requires the base subscription — there is no genuinely free tier anywhere in this list, only cheaper and more expensive ways to pay.² The full economics — including where that money actually goes and what it costs to run a season on each — sit on p15.

Le Mans Ultimate's position on this list deserves a fuller word than "financially fragile," because the story underneath is genuinely unusual: a critically strong, technically ambitious sim shipped by a

publisher whose own SEC-adjacent filings describe an \$86.3 million deficit and a market capitalisation that had fallen to roughly \$8.25 million — figures that would sink most consumer software businesses outright.⁴ Motorsport Games' stated strategy, per reporting on CEO Stephen Hood's public comments, has been aggressive cost-cutting while seeking an outside investor or buyer, with staff reductions concentrated among junior roles. None of that has visibly slowed the product itself — LMU shipped its full 1.0 release on schedule in July 2025 and has kept releasing paid DLC track packs through 2026 — but a reader deciding whether to build a season around this platform is, honestly, betting on a company whose own numbers say it may not exist in its current form a year from now.

Gran Turismo 7's commercial shape is the inverse problem: a healthy, well-resourced platform whose depth is deliberately capped rather than financially constrained. It was the UK's third-biggest PS5 launch behind Elden Ring and Horizon Forbidden West, and per one analysis is now the highest-grossing Gran Turismo title in the US despite ranking only fifth by units sold among GT games — evidence of a smaller, higher-spending player base consistent with its live-service microtransaction model rather than a one-time purchase treated as finished.⁵ Real racing drivers have been quoted, per BoxThisLap, saying the base physics "feel shockingly right" even as GTPlanet's own community keeps the "simcade" label alive — accessible and well-regarded are not the same claim as deep, and GT7 is a clean case of the first two without quite the third.

92%

DECLINE IN RFACITOR 2'S PEAK CONCURRENT STEAM PLAYERS SINCE NOV 2023

IN BRIEF — WHAT EACH PLATFORM COSTS TO START

iRacing \$124.50/yr subscription + \$11.95+ per car/track.

ACC / Le Mans Ultimate one-time purchase, \$30–45, optional DLC packs.

Gran Turismo 7 / Forza one-time console purchase, no ongoing content bill.

Automobilista 2 frequently discounted, list price unconfirmed this session.

THE COMMUNITY READ

"Sharply mixed" is how one gaming-opinion piece characterises current sentiment on rFactor 2 — physics and force feedback still praised by veterans, alongside frustration at slow development momentum as attention shifts to Le Mans Ultimate, built by the same studio on the same underlying engine.³

SOURCES 1 Community and platform-comparison research compiled for this edition, Aug 2026. 2 iRacing.com, series structure, via iRacing Support documentation. 3 Gaming-opinion coverage aggregated for this edition, treated as colour rather than hard fact; SteamDB/Tracker.gg for the numeric decline. 4 Insider Gaming; Overtake.gg; Motorsport Games SEC filing (EX-99.1) via StockTitan; Motorsport Week. 5 Wikipedia, "Gran Turismo 7"; BoxThisLap analysis; GTPlanet forum discussion.

THE GEAR, PART ONE

The Rig

From a \$299 bundle to a \$6,290 pedal, alone. Buy-once-cry-once, tier by tier.

Every price below is dated to August 2026 and sourced to a retailer, review outlet or the manufacturer's own listing as surfaced in this book's research; manufacturer sites themselves were largely unreachable from this session's network, so figures are cross-checked across independent listings rather than confirmed on primary product pages, and are flagged accordingly.¹

Entry (roughly \$300–500). Moza's R9 V3 bundle (wheelbase, wheel, pedals) has been seen at \$299; its R3 (3.9Nm) and R5 (5.5Nm) bundles list around \$379–399. Simagic's Alpha Mini (10Nm) lists near \$399. Fanatec's CSL DD base is \$349.99, with an \$8Nm Boost Kit sold separately at \$149.95 or bundled at \$479.95.²

Mid (\$700–1,200). This is where 2026's price-floor break landed: Moza's R21 Ultra (21Nm) at \$699 and R25 Ultra (25Nm, adding a hardware torque sensor) at \$899 — both using a new flat-wire, "zero-cogging" motor and a 21-bit magnetic encoder resolving roughly two million wheel positions, about ten times

its predecessor's resolution.³ Simagic's Alpha (15Nm) and Alpha U (23Nm) sit at \$649 and \$899. Simucube's 2 Sport (17Nm) has been seen around \$800–999. Fanatec's new flagship Podium DD (25Nm consistent, 33Nm peak, replacing the DD1/DD2 line) launched Q1 2026 at \$1,199.99 — deliberately priced below the model it replaces.⁴

The R21 Ultra specifically undercuts its own predecessor, the outgoing 2021 R21, which sold at \$849 — meaning Moza's new mid-tier flagship is \$150 cheaper than the model it replaces while gaining torque, encoder resolution and a redesigned motor. Reviewers noted a direct rival collision at the exact same price: Simagic's Alpha Evo Pro (18Nm) also lists at \$699, putting two serious direct-drive bases from two different manufacturers at an identical price point for the first time either brand has tested that exact number.⁷ The bigger disruption sits one tier up — the R25 Ultra's \$899 undercuts Simucube's established 25Nm benchmark, the 2 Pro at roughly \$1,499, by \$600, at matching peak torque.

TORQUE PER DOLLAR — 2026's PRICE-FLOOR BREAK

Wheelbase price (USD) vs. rated torque (Nm) — selected models, Aug 2026



Bar length ∝ price at roughly constant torque class (21–25Nm) — the R25 Ultra undercuts the Simucube 2 Pro by roughly \$600 at the same peak torque.

Sources: Traxion.gg, SimRacingCockpit.gg, Advanced SimRacing, ApevieSimulator — see sources, this page.

Fig. 2 — Selected 21–25Nm wheelbases, price vs. rated torque. Original figure, MERIDIAN, built from sourced listings.⁵

THE COMMUNITY READ

Reviewers called the R21 Ultra's pricing "a deliberate ploy by Moza to usurp its rival's flagship" — language strong enough that this book checked for an on-record response from Simucube or Simagic and found none. The silence is itself a data point about how the independents are choosing to fight this round.⁶

SOURCES 1 Retailer and press-listing aggregation compiled for this edition, Aug 2026; manufacturer sites unreachable this session. 2 Amazon/MOZA listings; simracewebshop.com; 6sigmasimracing.com; simhound.com; simagic.com listing snippets; GTPlanet (2021 launch price, reconfirmed); all via research compiled Aug 2026. 3 Traxion.gg, "MOZA R21 Ultra DD Wheel Base review"; SimRacingCockpit.gg; MySimRig; Bsimracing. 4 simagic.com listings; Slickdeals/Micro Center; Traxion.gg, "Fanatec launches flagship Podium DD"; OverTake.gg, "Fanatec Podium DD Review." 5 As sources 3–4, this page. 6 Traxion.gg, "MOZA R21 Ultra DD Wheel Base review." 7 Traxion.gg, "MOZA R21 Ultra DD Wheel Base review"; Advanced SimRacing, ApevieSimulator listings for Simucube 2 Pro pricing.

High (\$1,500–2,500). Asetek's SimSports Invicta (27Nm) has been seen around \$1,349 on sale. Simucube's 2 Pro (25Nm) lists near \$1,479. Fanatec's outgoing Podium DD2 (25Nm, being phased out for the new Podium DD) officially lists at \$1,599.99.¹

Over the top (\$3,000+). Simucube's flagship Ultimate (32Nm) has historically sat around \$2,000–2,500; this book found one unconfirmed reference to a "Simucube 3 Ultimate" at \$3,299–\$3,399 that could not be corroborated against Simucube's own site and should be treated as unverified pending confirmation. A "grail" setup — flagship base, a premium formula-style wheel rim (\$399–\$1,579 from Cube Controls), and Simucube's ActivePedal Ultimate (\$2,059–\$6,290 depending on configuration) — commonly crosses \$3,000–\$8,000 before a cockpit is even added.²

Pedals split the same tiers. Entry load-cell sets (Moza's CRP2, \$369) give way to Fanatec's ClubSport V3 (~\$400–470) and Simagic's P1000/P2000 (\$469–670) at mid tier; Heusinkveld's Sim Pedals Sprint (from €495) is the reference standard most enthusiasts benchmark against, with the Ultimate line (from €675) adding hydraulic dampers per pedal.³ Above passive load cells sit active pedals — Simucube's ActivePedal Pro and Ultimate, which motorise resistance to simulate ABS pulsing and lockup rather than just measuring force — running \$1,849 to \$6,290 depending on configuration.⁴ This book found no evidence Moza currently sells an

active-pedal equivalent; that specific claim should not be printed as fact.

Wheel rims — the actual grip a driver holds, separate from the motor spinning it — are their own sub-market almost every buyer's guide undercounts. A base bundle typically ships one general-purpose road-car rim; serious racers commonly own two or three interchangeable rims for different disciplines, a GT3-style rim with a small display and rotary encoders for endurance racing, a bare formula-style rim for open-wheel classes, a deep-dish rally wheel for rallycross titles. Cube Controls' formula rims, cited earlier at \$399–\$1,579 depending on model, sit at the accessible end of a market that runs considerably higher for licensed replica rims of specific real cars, a category this book's research could not price with confidence and does not attempt to.

A caution worth stating plainly before the ladder above reads as gospel: nearly every price in this section moves week to week, sales and bundle promotions are constant across this market, and a figure "seen" at one retailer on one day is not a standing price. Treat every number on this page and the one before it as a snapshot from early August 2026, not a current quote — the pattern of who undercuts whom, not the exact dollar figures, is the part of this section likely to still be true by the time this book is read.

IN BRIEF — WHAT A COMPLETE ENTRY SETUP ACTUALLY COSTS

Wheelbase + wheel + pedals bundle ~\$299–399 (Moza R9/R3/R5-class)

Add a load-cell pedal upgrade +\$99–369 (Moza CRP2 class)

Add a basic monitor stand or wheel-stand mount +\$100–200, no dedicated cockpit

Realistic all-in entry point roughly \$500–700 before a cockpit

THE CROSS-EXAMINATION

A "Simucube 3 Ultimate" at \$3,299–\$3,399 surfaced exactly once in this book's research, contradicting the stable "Simucube 2" naming used everywhere else — quite possibly a mis-scraped listing rather than a real product. It is named here only as a warning against treating a single search result as a fact, not as something to shop for.⁵

SOURCES 1 6sigmasimracing.com; Advanced SimRacing; ApevieSimulator; fanatec.com listing text via velocityprosims.com/simracereviews.com, all via research compiled Aug 2026. 2 WebSearch aggregation; unverified "Simucube 3 Ultimate" claim flagged explicitly rather than printed as confirmed; Cube Controls listings; simucube.com listing snippet. 3 simhound.com; simline.eu; simmotion.com; simracingsetup.com "Heusinkveld Ecosystem Explained." 4 simucube.com listing snippets; raceful.com review. 5 As source 2, this page.

THE GEAR, PART TWO

The Build

Where the "spare room" part of the hobby actually happens — and the tier of the market this book could not properly map.

Cockpits are the part of the hobby that most resembles furniture-building. At the accessible end, Trak Racer's RS6 Racing has been seen around \$424; Sim-Lab's GT1 Pro chassis (frame only — seat, mounts and pedal tray sold separately) starts near \$649, with the PIX Pro roughly \$150–200 above it.¹ A configured Trak Racer TR8 Pro build with seat and mounts has been reviewed as costing close to \$1,899, against a bare frame closer to \$600–700 — two very different numbers worth not confusing.²

The DIY tier runs on aluminium extrusion — 8020 or Misumi rail, bolted into a rig by the owner. Community consensus, per one 2026 buyer's guide, is unambiguous: "for most buyers, a pre-built aluminium-profile kit delivers the strongest balance of rigidity, modularity, time savings and predictable cost. DIY remains the right answer when the design must be genuinely custom."³ No reliable single "a DIY 8020 rig costs \$X" figure survived this book's research — total cost depends heavily on parts sourcing and whether the builder already owns tools, and this book declines to print a specific number it cannot stand behind. SimHub, the community's standard DIY dashboard and motion-control software, is free at its core, with paid add-ons unlocking its Motion module; it supports roughly 100 games and 200 devices, including no-code Arduino integration for shift lights and displays.⁴

Displays split along a real, well-evidenced line: triple monitors remain what most competitive racers use — full mirror visibility, sharp telemetry overlays, no motion sickness, no headset fatigue across a multi-hour stint — while VR (Meta Quest 3 as the accessible entry, Pimax Crystal Light/Super and the Dream Air at the high end, Valve Index now explicitly "not

recommended unless you already own one") wins on immersion and solves the physical-space problem entirely.⁵ Multiple independent buyer's guides converge on the same framing: triples for competitive league racing, VR for hobbyist immersion.

A specific VR entry worth naming because it complicates the "VR for hobbyists only" framing: Valve's Steam Frame, a streaming-first standalone headset running SteamOS with Wi-Fi 7, marks the company's return to VR hardware — but at least one reviewer surfaced in this book's research raised doubts about its suitability for sim racing specifically, given a design built around wireless game streaming rather than the low, consistent latency competitive racing needs. This book treats that as a forward-looking caution rather than a settled verdict, since the headset had not accumulated a real sim-racing track record at the time of writing.⁶

Chassis material matters more than marketing copy usually admits. Aluminium extrusion rigs flex measurably under hard braking loads compared to welded steel — a difference serious racers describe less as "better" or "worse" than as a trade-off between the extrusion rig's total modularity (any panel, any angle, reconfigurable in an afternoon) and a welded frame's stiffness once a driver's preferred geometry is actually settled. Sim-Lab and Trak Racer both sell into the extrusion camp; this book found no major manufacturer building serious welded-steel rigs at consumer volume, which may explain why the custom-maker tier discussed opposite is so thinly documented — steel fabrication at this scale appears to be a bespoke, one-off trade rather than a manufactured product category.

200

DEVICES SIMHUB SUPPORTS — SHIFT LIGHTS, DASHBOARDS, MOTION RIGS — FREE AT ITS CORE

SOURCES 1 simracingcockpit.gg "Sim-Lab Buyer's Guide 2026"; sim-lab.us listings; pure-sims.com; simracingsetup.com. 2 racingrigguide.com review, distinguishing frame-only from a fully configured build. 3 prosimhq.com, "8020 Aluminum Profile Sim Rigs: DIY vs Pre-Built (2026 Guide)"; raceanywhere.co.uk; OverTake.gg forum thread. 4 simhubdash.com official site text; tecnobits.com; simracingcockpit.gg. 5 simracingworks.com; simracingcockpit.gg "VR or Triple Monitors for Sim Racing?"; simrigzone.com; simracingsetup.com "The Best VR Headsets For Sim Racing in 2026." 6 simracingsetup.com "The Best VR Headsets For Sim Racing in 2026," on the Valve Steam Frame.

Motion platforms are the genuine over-the-top tier. Next Level Racing's Motion Platform V3 (two-actuator pitch/roll) is described across multiple retailers as "the most popular consumer option," listed from roughly \$2,999–3,499.¹ DOF Reality's P3 (three-axis) runs \$5,999–6,199; its P6 — true six-degrees-of-freedom motion, heave, roll, pitch, surge, sway and yaw — lists at \$9,999.² The clearest single "over the top" anchor this book found: Sim Coaches' Elite Motion Racing Simulator, a full 6DOF turnkey rig with seven actuators and triple 45-inch OLED displays, listed starting from \$49,995, delivered and installed.³

The custom-maker tier — small boutique builders enthusiasts commission from directly, the way a hi-fi obsessive commissions a cabinet-maker — is the one part of this gear landscape this book could not properly map. One named shop, Digital-Motorsports.com, describes itself as building custom rigs in-house with motorsport professionals, working with Sim-Lab and Heusinkveld components; this book could not independently corroborate that claim against a second source and prints it as a single-source lead, not a vetted recommendation.⁴ Etsy hosts handmade and 3D-printed sim racing components, but no named small-batch welded-steel or CNC boutique full-rig maker emerged from this book's research with the kind of press coverage a custom PC-case maker or a bespoke furniture builder would have. That is worth stating plainly rather than papering over: this corner of the hobby is either genuinely thin on documented names, or simply invisible to search — and a magazine that

claims to have mapped a community owes the reader that honesty over a confident-sounding but unverified name.

Motion is the tier of this gear ladder most likely to be genuinely misunderstood by an outsider, and the sceptic's case against it deserves a fair hearing here rather than being saved entirely for p23: a motion rig does not make a driver faster, and several sources in this book's research — including Max Verstappen's own comments on rig choice, quoted in full on p23 — describe motion platforms as measurably slower to drive on than a static rig, because the driver's own inner ear is now receiving cues the physics engine cannot fully control. What motion buys instead is a different kind of information: an early, physical warning of grip loss through the seat of the pants, the way a real car communicates before the eyes or hands do. That is a genuine, real thing to want from this hobby — it is simply not the same thing as lap time, and the two should never be confused when a \$50,000 rig is on the table.

Between the two extremes mapped across these four pages — a \$299 bundle and a \$50,000 turnkey motion studio — sits the actual median this book's research kept surfacing: a \$700–1,500 wheelbase, a \$400–600 pedal set, and a \$650–850 cockpit frame, landing a genuinely serious setup somewhere around \$2,000–3,000 all in, before any motion, VR headset or custom rim is added. That is the number a newcomer reading this book cover to cover should actually budget toward, not the headline extremes either end of the market is built to advertise.

\$49,995

STARTING PRICE, SIM COACHES' 7-ACTUATOR 6DOF TURNKEY MOTION RIG — DELIVERED & INSTALLED

IN BRIEF — THE CUSTOM TIER'S HONEST LIMITS

Digital-Motorsports.com single-source lead, not independently corroborated.

Etsy hosts handmade/3D-printed components, no named full-rig maker found.

Named boutique welders/CNC shops comparable to bespoke PC-case makers: none surfaced.

Read this as a genuinely thin, undocumented corner of the hobby — not a gap in the research.

THE CROSS-EXAMINATION

Buyer's guides routinely present DIY extrusion builds as the "authentic" path and pre-built kits as a compromise — but the actual community consensus found in this research runs the other way: a pre-built kit is now the *recommended default* for most buyers, DIY reserved for genuinely custom needs.⁵ The romantic version of this hobby is not always the one its own community actually recommends.

SOURCES 1 staples.com; billionaire toys.com; walmart.com; gigaparts.com; boostedmedia.net. 2 dofrealty.com official listings; apevie.com; amazon.com aggregation. 3 simcoaches.com product listing, Aug 2026. 4 digital-motorsports.com listing, single-source, not independently corroborated. 5 prosimhq.com, "8020 Aluminum Profile Sim Rigs: DIY vs Pre-Built (2026 Guide)."

INDUSTRY & ECONOMICS

The Consolidation

Corsair now owns the wheel and the seat. Whether that's good for the buyer depends entirely on who you ask.

Corsair Gaming's move into sim racing hardware began in earnest on 9 May 2024, when it announced exclusive negotiations to acquire the Fanatec product line from Endor AG, a German company then carrying roughly €70 million in debt. The deal closed that September — one search-derived summary places it 19 September, another, more likely correct given it matches the actual release URL, 23 September 2024. Then-CEO Andy Paul framed the rationale around Corsair's supply-chain scale helping the line "return to profitability."¹

Twenty-two months later, on 5 August 2026, Corsair announced it was buying Trak Racer — the Melbourne-founded maker of cockpits, seats and motion rigs Fanatec's own hardware bolts into — and folding it in as a Fanatec sub-brand under a new "SIM Solutions" group. Trade press cites cash consideration around \$40 million plus a \$10 million performance earn-out, roughly "\$50 million" combined; Corsair's own SEC 10-Q, filed for the quarter ended 30 June 2026, discloses the transaction as a subsequent event at approximately \$43.7 million net cash consideration — a figure this

book could not fully reconcile against the trade-press number without access to the full filing text, and prints both rather than picking one.² Trak Racer's founder, Matt Sten, joins as CTO of the new SIM Solutions group; Trak Racer's existing CEO, Pete Hosking, continues running day-to-day operations.³ Current CEO Thi La, who took over in a planned transition earlier in 2026, called it "a continuation of our strategy to scale our Gamer and Creator Peripherals segment into a much larger addressable market."⁴

The strategic logic, per trade coverage: Fanatec supplies the electronics — wheelbases, pedals, wheel rims, software — and Trak Racer supplies the structure underneath them, so Corsair can now sell one turnkey rig instead of two separate purchases from two separate vendors. Left unaddressed publicly, per the same coverage: Trak Racer had an existing bundling relationship with Simucube, and Simagic occupies roughly Fanatec's own market tier — an open question about what happens to those partnerships now that Trak Racer sits inside a Fanatec-branded group, one no source in this book's research has resolved.⁵

Two acquisitions, twenty-two months apart, one parent company.

FRAMING

CORSAIR'S OWN MATERIALS

A turnkey simulation platform, spanning racing and flight sim, with attractive margin-expansion opportunities and a meaningfully expanded product portfolio for Fanatec's ecosystem.⁶

FRAMING

THIS ISSUE'S OWN READING

Two of the hobby's four or five real hardware brands now answer to one Nasdaq-listed parent — real for buyers if it means one warranty line and one shipping account, and worth watching if it means one company setting the market's price umbrella instead of four competing on it.

THE COMMUNITY READ

Community reaction to the deal split cleanly along the same line as the framing above: buyers welcoming one warranty line and one shipping account for a bundled cockpit-plus-wheel setup, against longtime forum voices uneasy that Fanatec's own 2024 acquisition by Corsair has now repeated itself on a second brand in under two years.⁷

SOURCES 1 Corsair IR, "CORSAIR Pursuing an Acquisition of Fanatec," 9 May 2024; Corsair IR, "CORSAIR Set to Acquire the Fanatec Product Line," Sept 2024, via BusinessWire. 2 Traxion.gg, "Trak Racer to operate as a Fanatec sub-brand in \$50m Corsair deal," 5 Aug 2026; TechPowerUp, "Corsair Welcomes Trak Racer to the Fanatec Family"; dotesports.com; Corsair Gaming Inc. Form 10-Q, period ended 30 Jun 2026, via SEC EDGAR. 3 BoxThisLap, "Corsair Buys Trak Racer, Joining Fanatec Portfolio." 4 citybiz/GameSpace coverage of Corsair's Aug 2026 release, via research aggregation — quote not independently confirmed against Corsair's original text. 5 esportsadvocate.net, "Corsair Acquires Trak Racer Assets." 6 As source 4. 7 Community-forum sentiment aggregated for this edition, characterised rather than tied to one named post.

iRacing's own economics stack a subscription on top of per-item content purchases, in perpetuity — the structural point critics keep returning to on p21. From 18 August 2026, membership runs \$124.50/year (up from \$110, only the second increase since 2008), or \$14.75/month; additional cars cost \$11.95 each, tracks \$11.95–\$14.95, with volume discounts from 10% at three items to a permanent 20% past forty. One single-source estimate puts owning the entire current catalogue at roughly \$2,035 — a "buy everything" ceiling, not a typical spend; a realistic mixed road-and-oval roster sits well under that, in the low hundreds of dollars on top of the subscription.¹ Every one-time-purchase rival on p9 runs differently: Assetto Corsa EVO's early access costs \$39.99 outright; Le Mans Ultimate sells individual track-pack DLC at \$19.99 or a full season pass at \$44.99; ACC's GT-license packs run \$12.99 each. Own it once, own it outright — a structurally different economics story from iRacing's.²

For a Singapore-based buyer, the tax picture is simpler than the shipping one. Singapore charges no import duty on sim racing hardware at all — duty applies only to liquor, tobacco, motor vehicles and petroleum products — but GST, currently 9%, applies to essentially every import with no de-minimis exemption, calculated on cost plus insurance plus freight.³ A concrete shipping-cost figure for a wheelbase-sized parcel from the US or EU could not be sourced this session and is not printed as a guess. Local retail exists but is uneven by brand: Moza has genuine authorised local resellers (ElectronicsCrazy.sg, among others); Simagic similarly, via Simfai Solutions and S5 Tech. Fanatec's position is murkier — Fanatec's own support page states it has no designated reseller in Singapore or Indonesia, while at least one local retailer publicly claims "Authorized Premium Dealer" status with local

warranty — a live, unresolved contradiction this book flags rather than settles.⁴ Simucube appears to have the thinnest confirmed local presence of the four brands profiled in this issue, suggesting most Singapore buyers import it direct.

Working through the arithmetic honestly: a Singapore buyer landing a \$899 Moza R25 Ultra direct from the US pays roughly \$80 in GST alone (9% of the sticker price before shipping is even added), on top of whatever a courier charges to move a wheelbase-weight parcel across the Pacific — genuinely more than the \$150 gap Moza's own price-floor break was designed to close against Simucube's \$1,499 2 Pro. Buying the same wheelbase through a genuine local Moza reseller, where one exists, sidesteps the freight entirely and likely the GST timing as well, since the reseller's own import GST is already priced into local retail. The practical lesson for a Singapore reader: check for a local authorised reseller before assuming a US or EU listing price is the real cost of ownership — the gap this book's research kept finding between "list price" and "landed price" is often larger than the difference between two competing brands.

Market-size estimates for sim racing hardware vary widely by research firm and methodology — one 2026 estimate puts the "racing simulator market" at \$1.30 billion, projected to \$2.46 billion by 2034 at an 8.3% compound annual growth rate; a separate "sim racing hardware" report puts a 2025 baseline of \$2.28 billion growing to \$3.85 billion by 2033 at 6.77%.⁶ Neither number should be read as precise — commercial market-research reports of this kind are frequently commissioned by the industry itself and rarely disclose their full methodology — but both agree on the direction: real, sustained growth, which is exactly the kind of market a hardware company with spare acquisition capital goes looking for.

9%

SINGAPORE GST ON IMPORTED SIM GEAR — NO DUTY, NO DE-MINIMIS EXEMPTION

THE CROSS-EXAMINATION

Fanatec's own support page states it has no designated reseller in Singapore, while a local retailer publicly claims "Authorized Premium Dealer" status with local warranty on the same brand. Both cannot be fully true at once, and this book prints the contradiction rather than picking the more flattering version.⁵

SOURCES 1 BoxThisLap, "iRacing Subscription Price Increase, August 2026"; OverTake.gg; simracingsetup.com, "How Much Does All Content in iRacing Cost?" (single-source total estimate). 2 OverTake.gg, on Assetto Corsa Evo pricing; GameNGuide, on Le Mans Ultimate DLC; LMU Wiki DLC page; Steam listing, ACC GT World Challenge Pack. 3 IRAS, GST e-tax guide on the Overseas Vendor Registration regime; Singapore Customs, "Duties and Dutiable Goods Overview." 4 ElectronicsCrazy.sg; Simfai Solutions; S5 Tech listings; Fanatec Support, "Does Fanatec have reseller in Singapore or Indonesia." 5 As source 4, this page. 6 Market-research report aggregation compiled for this edition, Aug 2026 — figures unverified beyond the reports' own published summaries.

RACING & ESPORTS

The League

A safety-rating system built to automate stewarding across thousands of daily races, and the sim racers who turned a controller into an actual race seat.

1 Racing runs its competitive structure on two separate numbers, routinely confused by newcomers. iRating (iR) is an ELO-style skill rating drawn from race results; Safety Rating (SR) measures clean, consistent driving and is what actually promotes a driver up the license ladder — Rookie, then D, C, B, A, and finally Pro.¹ Rookie clears into D after two races or four time trials at SR above 3.0; each subsequent class needs Fast Track promotion at SR 4.0 or above; A-class drivers reach Pro by finishing top-12 in an A-series race within the discipline's first three seasons — a rule this book sources to a semi-official coaching site rather than iRacing's own current rulebook, and flags as needing direct confirmation.²

At the top sits real prize money. The eNASCAR Coca-Cola iRacing Series carries a \$500,000 prize pool, an actual NASCAR trophy and championship ring, and \$100,000 to the champion — Steven Wilson took the 2025 driver title, Spire Motorsports the teams' title.³ The Porsche TAG Heuer Esports Supercup, iRacing's single-make GT3 Cup series, has run a \$200,000 pool in both 2024 and 2025.⁴

Formula 1's own esports arm — launched 21 August 2017 as the F1 Esports Series, renamed F1 Sim Racing for the 2023-24 season — has run prize pools of \$700,000–\$750,000 and seen every major constructor field a contracted esports squad since 2018. Champions: Brendon Leigh (2017, 2018), David Tonizza (2019), Jarno Opmeer (2020, 2021, and again 2025 with a third constructor, Oracle Red Bull Sim

Racing — the first driver to win with three different teams), Lucas Blakeley (2022), and Frederik Rasmussen (2023-24).⁵

The inaugural 2017 season is worth pausing on for scale alone: over 60,000 players attempted to qualify for the live finals from a standing start, in a genre that had no official esports pathway at all the year before, and the broadcast reached viewers in 123 countries. 2018 marked the turn from open competition to a proper league structure — real constructors fielding contracted drivers for the first time, Mercedes-AMG Petronas Esports taking the inaugural teams' title with Leigh retaining his individual crown. That shift, from "anyone can enter" to "teams recruit and contract," is the same shift iRacing's own license ladder formalises through Safety Rating rather than an open qualifier, and it recurs across every serious esports structure profiled in this section.⁷

A newcomer's realistic path onto any of these ladders starts nowhere near a contract. iRacing's own structure makes this explicit: Rookie and D-class racing is genuinely open to anyone who pays the subscription, with promotion driven by clean, consistent racing rather than raw speed — meaning the actual bottleneck for most newcomers is temperament and patience, not talent. The drivers profiled opposite did not skip that stage; they simply climbed it fast enough, and publicly enough, to be noticed by an organisation with a real budget.

\$500,000

ENASCAR COCA-COLA IRACING SERIES PRIZE POOL, 2026 SEASON

THE COMMUNITY READ

Opmeer's real-world career ended for lack of sponsorship after Spanish F4 and Formula Renault Eurocup; sim racing gave him a second, better-paid career the real one never offered. He is not alone in that shape of story — the whole point of this section.⁶

SOURCES 1 Traxion.gg; iRacing Support documentation, on iRating vs. Safety Rating. 2 Coach Dave Academy; iRacing.com "Getting Started" — Pro-license criteria flagged as needing confirmation against iRacing's current official rulebook. 3 Jayski's NASCAR Silly Season Site; EsportsInsider; Liquipedia (via snippet). 4 iRacing.com press materials; Porsche racing site. 5 Autosport; Formula1.com; Wikipedia, "F1 Esports Series"; RaceFans; The Race, on Opmeer's real-racing background. 6 The Race, on Jarno Opmeer's real-racing career and its end. 7 Autosport; RaceFans, on the 2017–2018 F1 Esports Series seasons.

The endurance side runs through the Le Mans Virtual Series, a joint venture between Motorsport Games and the Automobile Club de l'Ouest — the real organiser of the 24 Hours of Le Mans — on the rFactor 2 platform, with a \$250,000 series prize fund across LMP and LMGTE classes.¹ Its 2023 Le Mans Virtual 24 Hours saw #2 Team Redline take the LMP class with a lineup including Felipe Drugovich (2022 Formula 2 champion) and Felix Rosenqvist (IndyCar racer, ex-F1 test driver); R8G Esports, founded by then-Haas F1 driver Romain Grosjean, won GTE.² The series has also produced its low point on record: at the January 2023 finale, leaked IP addresses caused mass disconnections mid-broadcast; Max Verstappen, then leading the standings, dropped to 17th, hit further connection issues, and quit live on stream, calling the organisers' handling "incompetence" and the event "a clown show," on the record stating it would be the last time he raced in the series.³

The sim-to-real pathway is real but rare, and its clearest case is Igor Fraga: a genuine Brazilian racing career (Brazilian F3 champion 2017, USF2000 rookie season 2018) running in parallel with sim racing success, including winning McLaren's Shadow Project talent search. In 2020 he won the Toyota Racing Series in New Zealand outright — beating future F1 drivers Liam Lawson, Yuki Tsunoda and Franco Colapinto to the title — and by 2025 was a

Super Formula rookie in Japan with Nakajima Racing.⁴ Rudy van Buren's 2017 win of McLaren's earlier "World's Fastest Gamer" competition made him McLaren's official simulator driver, appearing alongside Fernando Alonso and Stoffel Vandoorne at the team's 2018 car launch.⁵ Chris Lulham, a UK Team Redline sim racer with wins at the iRacing Nürburgring 24 Hours and two VCO Esports World Cup titles, switched fully to real racing in 2024, won the Radical Cup UK outright, and by 2025 was racing GT World Challenge Europe for Emil Frey Racing — a seat that opened through Max Verstappen's own driver programme.⁶

Enzo Bonito's story sits between the two extremes and is worth telling for its shape rather than its ending: an iRacing GT3 season winner and Team Redline member who later joined Ferrari's driver-development orbit as a Pro Exhibition esports driver, Bonito won the first-ever eROC (esports Race of Champions) World Final, earning a start in the real 2019 Race of Champions in Mexico City — where he beat 2016-17 Formula E champion Lucas di Grassi and Indy 500 winner Ryan Hunter-Reay in actual cars, on an actual track, having qualified for the seat entirely through simulator results. It is one of the cleanest single data points this book found for the pathway working exactly as advertised, even without a subsequent full-time racing career attached.⁸

IN BRIEF — NAMED ESPORTS TEAMS

Williams Sim Racing founded 2018, first F1 team with a dedicated esports arm; 35+ drivers across ~10 titles.

Oracle Red Bull Sim Racing founded 2018, seven world titles across disciplines; partnered with Team Redline since Oct 2023.

Porsche Coanda Esports German org, founded 2014, leading force in iRacing/Le Mans Virtual GT racing.

R8G Esports founded April 2020 by Romain Grosjean; runs a Praga Karts real-world pathway programme.

THE CROSS-EXAMINATION

The sim-to-real pathway is real but survivorship-biased in how it gets told: Fraga, van Buren and Lulham are the names that made it, cited repeatedly because they're the ones who did. This book found no data on how many McLaren Shadow Project or comparable entrants tried and simply didn't — the success stories are the whole visible record, which is worth remembering before treating the pathway as reliably open.⁷

SOURCES 1 24h-lemans.com; Motorsport Games press releases; InsideEVs. 2 Traxion.gg headline coverage, 24 Hours of Le Mans Virtual 2023 results. 3 The Drive; PC Gamer; NME; Motor Sport Magazine; RacingNews365, on the Jan 2023 Le Mans Virtual finale. 4 Formula Scout; Toyota NZ; honda.racing, on Igor Fraga's career. 5 ESPN; pressreader/Arab Times; McLaren.com, on Rudy van Buren. 6 Autosport (snippet-only, full text unreachable); corroborated via Team Redline/GT World Challenge Europe result listings. 7 Editorial assessment, this edition, on survivorship bias in sim-to-real coverage. 8 Supercars.net; SCMP; Ferrari.com, on Enzo Bonito's eROC/Race of Champions result.

THE COMMUNITY MAP

The Community

Mapped honestly, including the names this book could not confirm and the ones it deliberately left out.

r/simracing is the hobby's largest public forum, aggregator-estimated near 375,000 members — an approximate, single-source figure this book cannot independently confirm and prints as such.¹ OverTake.gg is the other major hub: it began as RaceDepartment, an independent forum and mod-hosting site founded by Bram Hengeveld in 2006, and merged into OverTake.GG — a German site operated by Freaks 4U Gaming with Porsche funding, launched 2020 — in early 2023, folding RaceDepartment's forums and mod library into the combined brand.² Boosted Media, run by a creator working under the name Jack, is a hardware-review outlet rather than a forum — independent wheelbase, pedal and cockpit reviews plus a showcase triple-65-inch motion rig used for testing.³

Named, verifiable creators include Jimmy Broadbent — a UK creator with over a million subscribers since 2012, who crossed into real motorsport with a British Endurance Championship class win and now races the Nürburgring Langstrecken-Serie for Team Fanatec — and Lucas Blakeley, the 2022 F1 Esports champion who parlayed the title into a real GB4 seat with a record winning margin.⁴ Sim Racing Garage, hosted by Barry Rowland, centres on building high-end rigs and studio setups. This book also tried, and failed, to verify several names sometimes attached to the scene — a "WirthTV" or "JJ Wirth," a "VR Flavio," a specific "Josh" at OverTake — and found no credible

match for any of them. They are named here only to be struck, not repeated as fact.⁵

Real-world venues exist but their status is genuinely uncertain in places. RaceSim1 in Toronto — five cockpits, 49-inch screens, laser-scanned tracks — is listed as permanently closed on one review platform and as currently operating on three others; this book could not resolve the contradiction and declines to state it is open. Singapore's own venue, Legion of Racers, is the one exception with a clean, corroborated record — its full profile runs in the Long Read, p25.⁶

A structural reason this community is harder to map than most hobbies this book has covered: it splits cleanly across three separate platforms with almost no crossover audience. iRacing's own official forums serve the subscription racer, deliberately walled off and moderated to that community's own norms; r/simracing and OverTake.gg serve the platform-agnostic hardware and setup crowd who may never touch iRacing at all; and YouTube serves a third audience again, watching build videos and lap-time comparisons who may never post on either forum. A creator can be genuinely enormous within one of these three worlds and functionally invisible in the other two — which is very likely why several names this book tried to verify (see the struck names above) surfaced confidently in one search context and nowhere else.

375,000

APPROXIMATE R/SIMRACING MEMBERSHIP — AGGREGATOR-ESTIMATED, SINGLE-SOURCE

THE CROSS-EXAMINATION

A hobby press this online-native produces an unusual amount of confident-sounding noise — plausible creator names, plausible venue names, plausible member counts, all repeated until they read as fact. This section's refusals (the unconfirmed creators, RaceSim1's open/closed contradiction) are as much the point of a community map as the confirmations are.⁷

SOURCES 1 gummysearch.com/r/simracing; simracelounge.com, single-source approximate figure. 2 overtake.gg/news, "Better Together" and "Welcome to the new look RaceDepartment" posts; Traxion.gg, "Porsche and Overtake.GG purchase sim racing community RaceDepartment." 3 boostedmedia.net; YouTube channel description; au.youtubers.me. 4 Wikipedia, Jimmy Broadbent; Autocar; Wikipedia, Lucas Blakeley; McLaren's own site. 5 Search research compiled for this edition, Aug 2026 — no credible match found for any of the three names. 6 racesim1.com; Yelp ("CLOSED - Updated March 2026"); Tripadvisor; Facebook — status contradictory across sources. 7 Editorial assessment, this edition.

Discord activity is real but its numbers move constantly — treat all of the following as approximate snapshots, not stable counts. EuroRacers, focused on Assetto Corsa/ACC/Automobilista 2 content creation, was the largest server found, around 58,300 members; "That Sim Racing Discord" around 17,500; r/simracing's own official server around 9,300.¹

Nils Naujoks is a real, verifiable competitive sim racer — a BS+Competition driver who won the 2022 BMW Sim Time Attack in ACC — but this book found no evidence supporting a "journalist" framing sometimes attached to his name, and does not repeat that specific claim.² This kind of small, specific correction matters more than it looks: a community map that can't tell a competitive driver from a journalist has not actually done the reading.

The commercial rig-rental sub-industry — companies like The Carnival Fair, FunCo, EventGuru and Solve Entertainment renting F1-branded rigs for parties and corporate events — is worth a line precisely because it is not the same thing as the hobbyist scene above it, even though it uses identical hardware. Confusing "a rig you can rent for an afternoon" with "a community that races weekly" is a common category error in coverage of this hobby, and this book takes care not to make it.³

A name that surfaced only once, weakly sourced, and is printed here purely as an example of how this book handles a thin lead rather than as a confirmed fact: an "Aarava" appears in one reference to a "Racing Prodigy Search Committee," a talent-scouting body said to evaluate sim racers for esports pathways. This

book could not confirm a channel, a subscriber count, or even a full name behind the reference before its research budget closed, and does not build any claim on top of it. It stays in this book for exactly one reason — to show the difference between a name this book struck outright (the unconfirmed creators on the previous page) and a name it flags as genuinely uncertain rather than false. Both outcomes look similar to a reader; they are not the same finding.⁴

Taken as a whole, this community reads less like a single hobby and more like three overlapping ones sharing hardware: a competitive-racing culture centred on iRacing's ladder and its official series (p17), a build-and-review culture centred on YouTube and Boosted Media, and a forum culture — r/simracing, OverTake.gg, the Discord servers above — that argues about both. The fights that culture actually has, rather than the platforms it argues on, are the subject of the next section.

One more honest gap before this book moves on: age and gender demographics for this community were not something this book's research could source with any confidence at all — no survey, no platform-disclosed breakdown, nothing beyond the visual impression of who appears in event photography and creator content, which this book does not treat as data. Sim racing's own community forums occasionally discuss this openly, usually in the context of wanting the sport to broaden, but this book found no hard numbers behind those conversations and declines to guess at a demographic picture it cannot actually support.

IN BRIEF — NAMED CREATORS, CONFIRMED VS. STRUCK

Confirmed Jimmy Broadbent, Lucas Blakeley, Barry Rowland (Sim Racing Garage), Nils Naujoks (driver, not journalist).

Struck "WirthTV"/"JJ Wirth," "VR Flavio," a specific "Josh" at OverTake — no credible match found.

Flagged, not struck "Aarava"/Racing Prodigy Search Committee — one weak reference, not built on.

Method every name in this book was checked, not assumed, before printing.

A community map that can't tell a driver from a journalist hasn't done the reading.

2006

YEAR RACEDEPARTMENT — NOW OVERTAKE.GG — WAS FOUNDED

SOURCES 1 discord.com server listings/disboard.org tag directory, aggregated, approximate point-in-time figures. 2 Liquipedia Sim Racing wiki; podcast-guest appearance search, "journalist" framing not corroborated. 3 Event-rental industry listings compiled for this edition, Aug 2026. 4 Single search reference, uncorroborated, printed only to illustrate this book's verification standard.

CULTURE

The Wars

A subscription model critics call pay-to-race, a safety system built to automate stewarding at scale, and the scandals that made both necessary.

"Paying for everything but owning nothing" is a real, recurring line of complaint against iRacing's stacked subscription-plus-content economics on community forums — the full numbers run on p15. The counter-argument surfaces in the same threads: the cost functions as a quality filter, keeping a more invested driver base engaged and funding the laser-scanned content and heavier anti-grief infrastructure a genuinely free platform couldn't sustain. One data point supporting the counter-case: iRacing's most-populated series, the Mazda MX-5 Cup, runs entirely on content bundled with the base subscription — a real low-cost entry path exists inside a system its critics call pay-to-race.¹

iRacing's Safety Rating exists explicitly because online racing without it turns feral — the company's own published position is that a no-fault, automatically-counted incident system is "the only statistically viable way" to steward thousands of simultaneous daily races without human officials.² The scale of the problem it addresses is not abstract: NASCAR driver Hailie Deegan used a slur live on a January 2021 eNASCAR broadcast after contact with another driver, apologised publicly, and drew a concerned statement from sponsor Ford — a well-documented, multi-outlet-confirmed incident.³

Cheating scandals are this culture's richest, best-documented fight. In May 2020, Formula E driver Daniel Abt had 18-year-old esports racer Lorenz Hoerzing drive in his place, in disguise, during a COVID-era "Race at Home Challenge" — discovered via IP-address investigation, resulting in a €10,000 fine and Audi suspending his factory drive with immediate effect, ending his career there.⁴ At the 2022 iRacing Spa 24 Hours, drivers were caught

"grass dipping" — deliberately running two wheels off-track onto the verge to artificially cool tyre temperature for a lap-time gain; team URANO eSports withdrew in protest rather than use what it called "an exploit in the sim," and iRacing patched the exploit within days.⁵

The Abt case is worth sitting with a moment longer than the headline allows, because it is the rare sim racing scandal with a real-world career attached to the punishment. Abt was not an esports specialist trading on anonymity — he was a genuine Formula E factory driver, under contract to Audi, competing in what was explicitly billed as a real-drivers-only charity event during a period when actual racing had stopped entirely because of the pandemic. Hiring a ringer to preserve his standing in an event with no prize money at stake, only reputation, cost him the one thing that did have real money attached: his factory seat. That asymmetry — a low-stakes event, a career-ending consequence — is precisely the kind of story that makes the legitimacy debate on the next page harder to dismiss either way.

A parallel exploit surfaced in Gran Turismo's own esports scene: top players discovered that deliberately clipping a wall immediately after being bumped by another car tricked the game's automated penalty system into reading the contact as accidental rather than a track-limits violation, letting a driver "wallride" corners without penalty. The exploit spread unevenly by region — used openly in some Asia-Pacific competitions before being adopted reactively elsewhere, creating real confusion about what was and wasn't legal within the same season — until Polyphony Digital acted against it directly.⁷

THE COMMUNITY READ

At the January 2023 iRacing Daytona 24 Hours, Williams Esports' pole lap used an off-limits oval apron after the sim's track-limit detection had been manually disabled to let damaged cars limp home — Williams publicly apologised afterward, acknowledging the conduct "breached our standard for acceptable team conduct."⁶

SOURCES 1 Steam Community discussion threads, App 266410 (iRacing) — community forum sentiment, anecdotal rather than surveyed. 2 iRacing.com official article, "Safety Ratings — a cure for the mayhem in online racing games?"; iRacing support docs; BoxThisLap explainer. 3 CBS Sports; NBC Sports; 247Sports; Sportskeeda; Wikipedia, "2020 eNASCAR iRacing Pro Invitational Series." 4 CNN; ESPN; The Race; Motor Sport Magazine; Fox News; BBC Newsround, on the Daniel Abt/Lorenz Hoerzing scandal. 5 Traxion.gg; Race Sim Central; iRacing official patch notes, Season 3 Patch 3 Hotfix 1, Aug 2022. 6 OverTake.gg; PlanetFL; RaceFans; Esports News UK; zleague.gg, on the Jan 2023 Williams Esports incident. 7 OverTake.gg news coverage of the Gran Turismo wallriding exploit (search-result summary; full article unreachable this session).

F1 esports has its own, murkier cheating fight: allegations that some drivers used software (JoyToKey) or physical "grip sheets" to gain traction. Jamie Mels was disqualified by organiser PSGL after suspicious footage was flagged by a rival driver; Thomas Ronhaar and Alvaro Carreton were both separately accused, with Carreton accidentally showing a folder labelled "F1 grip hacks" live on stream, which he said was mod-testing for the game's publisher. Codemasters/EA has stated its own testing found no advantage from the alleged exploit; PSGL organisers have said publicly they believe further undetected cheating exists but is "nearly impossible to detect" without dedicated anti-cheat tooling — a structural gap named across multiple independent outlets as endemic to sim esports generally, since most titles were never built with anti-cheat systems at all.¹

The legitimacy question — is this a real sport — resolves less cleanly than either side wants. This book found no strong, named, on-record quote from a real driver flatly dismissing sim racing; what it found instead was a pattern of grudging convergence. Gran Turismo's own GT Academy, launched 2008 by creator Kazunori Yamauchi, set out specifically to convert top sim racers into licensed real-world drivers — proof of concept the genre has cited ever since. Max Verstappen owns a stake in Team Redline and streams iRacing seriously, past 2am after real qualifying sessions, while separately dismissing one specific arcade-style esports format as "just a game" — a real driver holding both positions at once, not a contradiction so much as a distinction critics of the "is it a sport" framing tend to flatten.²

This book found no citable, sourced thread documenting the hardware-elitism fault line — budget belt-drive wheels versus big-spender direct-

drive rigs — that this hobby's own participants describe as a live tension. It is real enough that entry-level brands market explicitly against it (Asetek's budget "Initium" line, covered by Notebookcheck as a deliberate lower-cost alternative to a premium-dominated market), but this book declines to invent a viral thread to illustrate a debate the evidence otherwise supports only structurally. That exact question — whether the spending actually buys anything — gets the full, honest test next, on p23.

Set the four fights on these two pages side by side and a pattern emerges that no single scandal shows on its own: every one of them is a dispute about fairness in a sport with almost no independent referee. iRacing's Safety Rating is the platform policing itself. PSGL's own admission that undetected cheating "is nearly impossible to detect" is an organiser policing a system it knows is porous. The Abt case was caught by amateur IP-address sleuthing, not an anti-cheat system. Real motorsport solved this problem over a century ago with stewards, scrutineers and a sanctioning body with legal teeth; sim racing, still young enough that most of its titles were never built with cheating prevention in mind at all, is visibly still working it out in public, scandal by scandal, patch by patch.

None of that is a reason to dismiss the culture as unserious — if anything the opposite. A community that publicly disqualifies its own drivers, patches its own exploits within days, and lets a team like Williams apologise on the record for its own conduct is a community holding itself to a real standard, imperfectly enforced. The absence of a century-old governing body is a genuine structural weakness; it is not the same thing as an absence of standards, and this book found no evidence the culture treats its own rules as optional once they exist.

€10,000

FINE THAT ENDED DANIEL ABT'S FORMULA E FACTORY DRIVE, MAY 2020

THE CROSS-EXAMINATION

PSGL's own organisers admit further cheating "is nearly impossible to detect" without dedicated anti-cheat tooling — meaning every scandal this book could name is, by the organisers' own logic, only the fraction that got caught. That is a genuinely uncomfortable note to end a culture section on, and this book prints it rather than softening it.³

SOURCES 1 Traxion.gg; Dexerto; EssentiallySports; EA Forums threads, on F1 esports cheating allegations; esports.net, "A Look into the Sim Racing Esports Cheating Pandemic"; SI.com, "Esports Has A Cheating Problem." 2 driver61.com/Sportskeeda, on Verstappen's rig comments; GT Academy history via multiple sim-racing press retrospectives. 3 As source 1, this page.

THE CROSS-EXAMINATION

The Sceptic

No. 66's *Rabbit Hole* ended on a one-line jab. This is the honest version of that argument, both sides taken seriously.

The claim, stated plainly: a \$2,000 direct-drive base does not lap a mid-pack racer faster than a \$500 belt-driven wheel — seat time and technique dominate hardware quality once a basic floor is cleared. The evidence for it is real but thinner than the hobby's own folklore suggests. Hardware-comparison content testing a Logitech G29 against a premium direct-drive wheel found gains described as "real but modest," not dramatic given the price gap, with three to five sessions needed before a driver fully adapts to new hardware at all — meaning some of the apparent gain from expensive gear is really just an adaptation cost the driver would have paid on any new wheel.¹ Coaching-content consensus, including from at least one hardware vendor's own blog being unusually candid about its own product category, converges on the same point: "high-end gear won't make you faster on its own" — reliability, adjustability and immersion support harder training, but don't substitute for it.²

The strongest evidence on record actually cuts the other way, and narrowly. A study on PMC, apparently published in *PLoS ONE* in January 2024, tested twenty experienced sim racers on load-cell versus Hall-effect brake pedals and found faster lap times and more consistent, "faster-driver-like" braking behaviour on the load cell — genuine, if this book's inability to open the primary text directly is set aside, evidence that pedal precision specifically does buy

real pace, distinct from the wheelbase question entirely.³ No comparable study exists, in this book's research, for wheelbases themselves.

On the "diminishing returns" question — is there a sweet spot where more money stops buying more pace — the answer found in aggregated buyer's-guide consensus (not one clear named source, closer to community folklore with a number attached) is a wheelbase around 8–9Nm of torque, already exceeding a real GT3 car's roughly 10Nm force at the wheel, delivering something close to the full experience.⁴ This book treats that figure as directionally credible and specifically unverified — printed with that caveat rather than as fact.

Worth naming explicitly what this claim is actually testing, because it is narrower than the way it usually gets argued online. "Does expensive hardware make you faster" collapses two different questions into one: does it make a given driver faster than their own baseline on cheap gear (almost certainly yes, marginally, per the evidence above), and does it make that driver faster than an equally experienced driver on cheap gear (the evidence found no support for this at all). The *Rabbit Hole*'s original one-line jab was really making the second, harder claim — that spending doesn't buy a competitive edge over an equally committed rival — and that is the claim this book's research actually supports, not the weaker first one hardware marketing prefers to imply.

20

EXPERIENCED SIM RACERS IN THE ONE STUDY THIS BOOK COULD FIND ON LOAD-CELL PERFORMANCE

THE COMMUNITY READ

Max Verstappen, on rig choice, is unusually direct: "I'm driving at home on a static rig because it's faster. All these motion rigs are slower. But of course, if you want to try and replicate a little bit of what you feel in real life, then yeah, you buy a motion rig." A real driver choosing the cheaper, simpler setup because it is objectively quicker — motion, not money, being the trade-off he names.⁵

SOURCES 1 DG Edge, "Does Expensive Sim Racing Wheels Make You Faster?," undated, 2026 context — summary-level source, primary text unreachable this session. 2 BoxThisLap, "Does a Better Sim Racing Wheel Translate to Faster Laps?"; Simucube blog, coaching-content aggregation. 3 PMC (PMC12251626)/DOAJ index entry consistent with a *PLoS ONE*, Jan 2024 paper — see also p8; primary text unverified this session. 4 Buyer's-guide aggregation compiled for this edition, unattributed to one clear named source. 5 driver61.com/Sportskeeda, quoting Max Verstappen — original interview date not independently confirmed.

The counter-case for spending is real, just narrower than the marketing around it. Better force feedback lets a driver feel grip loss and understeer onset earlier, which plausibly buys consistency — fewer offs, fewer spins — even where it does not obviously buy peak pace; this is a widely repeated mechanism across hardware-vendor and buyer's-guide content, plausible and consistent, but this book found no independent study measuring the effect specifically, as distinct from the pedal study above.¹ Two specific numbers surfaced repeatedly in pedal-vendor marketing content — a "0.5 seconds per lap" gain "in the first week" and "12% more accuracy in braking modulation" — trace to no named study or author in any source this book checked, and read as SEO copy borrowing credibility from the genuine PMC pedal study without being that study. Neither number appears anywhere in this book as fact.²

On whether the field's fastest drivers actually run the most expensive gear — the closest thing to a direct test of the original claim — no credible survey or dataset exists. The scattered evidence available cuts, if anything, against the premium-hardware thesis: at least two named esports pros have been reported using mid-range rather than flagship Fanatec equipment. That absence is itself worth printing as a finding: if elite performance clearly required \$2,000-plus hardware, the industry — which has every commercial incentive to publish exactly that correlation — would very likely have done so already, and it has not.³

The honest synthesis, holding both sides: skill and seat time dominate at the low-to-mid end of the

spending curve, a roughly \$500–800 setup captures most of the achievable lap-time benefit for a serious hobbyist, and money past that point buys consistency and comfort more reliably than it buys raw pace — with the one clear exception, load-cell pedals, where the evidence for a real, if modest, performance gain is genuinely the strongest in this entire book.

What money reliably does buy, even where it fails to buy lap time, is worth stating without cynicism: durability under a season of hard use, a warranty that matters when a motor genuinely fails, comfort across a three-hour endurance stint that a \$299 pedal set will not survive without recalibration, and — the one nobody puts in marketing copy — the simple pleasure of owning something well made. None of that is nothing. It is simply not the thing the original claim was arguing about, and a hobby this honest with itself should be able to hold "it won't make you faster" and "it might still be worth it" as two true statements at once, rather than needing the first to be false to justify the second.

If this book's own research had to be summarised in a single instruction for a newcomer standing at the start of p11's buyer's ladder: spend to the \$500–800 floor without guilt, spend past it only for reasons that are not about winning, and put whatever money is left over into seat time — a subscription, a season of races, a coach's hour of telemetry review — because every scrap of real evidence this book could find says that is where the actual speed lives.

\$500–800

WHERE COMMUNITY CONSENSUS PLACES THE REAL DIMINISHING-RETURNS FLOOR

THE CROSS-EXAMINATION, TURNED ON ITSELF

This section's own evidence base is thinner than its confidence should suggest — a handful of aggregator summaries, one unconfirmed primary study, and an absence of data standing in for a finding. That is not a reason to distrust the synthesis above; it is a reason to hold it exactly as tightly as the evidence supports, and not one notch tighter.⁴

SOURCES 1 Wikipedia, "Direct-drive wheel"; buyer's-guide and hardware-vendor content aggregation, plausible mechanism, no independent study located. 2 Apex Sim Racing; Apevie — both pedal vendors, figures uncited to any named study. 3 Traxion.gg, "Cop the set-up: the sim racing equipment the pros use" (title only, primary text unreachable); scattered single-driver equipment mentions, insufficient sample to generalise. 4 Editorial assessment, this edition.

THE LONG READ — UNDERCURRENT

The Studio at CT Hub

Singapore's sim-racing scene is small, real, and has already produced two national esports titles — mined the way this magazine mines any live regional story, not summarised from a press release.

Take the lift to the sixth floor of CT Hub, 2 Kallang Avenue, on a Tuesday afternoon, and you will find a walk-in sim racing studio charging from S\$25 an hour, open Tuesday to Sunday, closed Mondays.¹ It calls itself Legion of Racers, and its own marketing language — "Singapore's premier sim racing organisation," "2x GTWC Asia Esports champions" — reads, on its face, like the kind of self-description this book usually discounts. In this case the record backs it.²

Dillan Tan, a Singaporean racing for the Legion of Racers team, won the GT World Challenge Esports Asia Sprint Series in both 2022 and 2023, retaining the title in 2023 by winning four of five races.³ The team's second national title came through Indonesian driver Andika Rama Maulana — racing as "Ramstig" — who won the Toyota Gazoo Racing Asia Esports GT Championship in both 2023 and again in 2025, his second win coming off two race wins and a second-place finish across three rounds. Maulana's own account of his path, per an Indonesian esports profile, is unglamorous in exactly the way that makes it credible: he returned "seriously" to sim racing in 2018 specifically because a real racing career required money he didn't have, after playing community leagues since childhood.⁴ Three Legion of Racers sim racers also finished in the top five at the Asia Pacific GT E-Motorsport Championship in Hong Kong in December 2023, per Singapore outlet The Monitor SG.⁵

The organisation runs two divisions — LOR Racing for competitive tournament play, LOR Ascent for talent development — and describes a stated "virtual-to-real" pathway into karting through a sister team, structurally identical in ambition to McLaren's

Shadow Project or Gran Turismo's GT Academy, just built at Singapore's own scale rather than a global manufacturer's.⁶ A second brand, "Race Box," lists an identical CT Hub address and identical contact details to Legion of Racers itself — almost certainly a walk-in consumer booking sub-brand of the same operation rather than a genuinely separate venue, though this book could not confirm that on the ground and declines to state it as settled fact.

What makes Legion of Racers worth an Undercurrent rather than a paragraph is the same thing that makes any regional story worth mining properly: it is not a Singapore franchise of something bigger, it is a Singapore-built operation that went out and won against the wider region on its own terms, twice, in two different manufacturer-backed esports championships. Dillan Tan's 2023 title defence — four wins from five races — is not a marginal result; it is the kind of dominant season a team either builds a reputation on or gets accused of exaggerating, and the corroborating detail (races, dates, a named rival series) is specific enough that this book treats it as solid rather than promotional.

Andika Rama Maulana's presence on a Singapore-based team, despite being Indonesian himself, is its own small data point about how genuinely regional this scene already is beneath the national branding — a driver commutes to whichever organisation offers the best coaching and platform, not necessarily the one in his own passport's country, exactly as a real motorsport driver would. That is a maturity marker most fledgling esports scenes take years to reach, and Legion of Racers appears to have reached it already.

2x

NATIONAL GTWC/TGR ASIA ESPORTS TITLES WON BY LEGION OF RACERS DRIVERS

SOURCES 1 legionofracers.com, venue and pricing pages. 2 legionofracers.com/about-legion-of-racers. 3 legionofracers.com/news/dillan-tan-is-the-2023-gt-world-challenge-asia-esports-champion. 4 legionofracers.com/news/andika-rama-maulana-wins-the-tgr-asia-esports-gt-championship-2025; tribune.net.ph, 25 Nov 2025; ggwp.id, Andika Rama Maulana profile. 5 themonitor.sg, 19 Dec 2023. 6 legionofracers.com/about-legion-of-racers.

A second, membership-based body runs alongside the commercial studio: the Sim Racing Association (Singapore), operating as simracing.sg, describing itself as "Home of Singapore eMotorsport" and dating its own origins to 2011. Full voting membership costs S\$24 a year for citizens and permanent residents aged 21 and over; a free non-voting tier covers everyone else. Its leagues include an iRacing GT3 series run in partnership with Next Level Racing, a Macau-China sim racing association, and — a real, checkable institutional link — Motor Sports Singapore, the national motorsport federation.¹ One search summary suggested a further partnership with the Singapore Esports Association and IMG; this book could not independently confirm that specific claim against a direct source and does not repeat it as fact.

Set against the region, Singapore's scene is real but not the most institutionally developed in Southeast Asia. Malaysia's is: the Toyota GR Malaysia eRacing Championship is jointly sanctioned by the Motorsports Association of Malaysia and the Malaysia Esports Federation across three formal tiers, a level of federation integration this book found no equivalent of in Singapore.² Indonesia has produced the region's standout individual talent in Maulana. Australia sits a tier above both — Motorsport Australia, the national federation, runs its own dedicated "Esports Communities & Clubs" hub, and Supercars ran an official iRacing eseries around Bathurst during the COVID era — genuine federation-level sanctioning that nothing in Singapore, Malaysia or Thailand quite matches.³ Thailand's scene, by contrast, reads as community-only: SpeedR eSports and a Thai-language "Sim Racer Thailand" Facebook group exist, but this book found

no evidence of a professional-tier circuit distinct from commercial rig-building businesses — worth stating plainly rather than inflating a thin scene into a story it isn't.

What this book could not do — and says so rather than faking it — is find a single independently-sourced sentiment quote from a named Singapore sim racer describing what the scene actually feels like to be inside. Every warm description of Legion of Racers' studio traced back to an AI-summarised search snippet, not a page this book actually read and can stand behind; the honest version of this story leans on results and dates, which are solid, rather than atmosphere, which isn't yet sourced to this book's own satisfaction. That is the real shape of Singapore's sim-racing scene in August 2026: small, genuinely competitive, internationally credentialed at the podium — and still waiting for someone to properly write down what it feels like on a Tuesday afternoon at CT Hub.

Japan sits oddly outside this comparison entirely — home to Polyphony Digital and the Gran Turismo World Series, a genuinely global esports property that staged a 12-player offline showmatch in Japan in February 2026, yet this book found no named domestic Japanese club, forum or physical venue to set alongside Legion of Racers or Malaysia's federation-run league.⁵ That is likely a research gap rather than an actual absence — a hobby this dependent on a shared language for its forums and Discords is exactly the kind of scene a search conducted in English will systematically undercount — and this book names the gap rather than pretending Japan simply has no domestic scene worth mentioning.

IN BRIEF — THE REGION, COMPARED

Singapore Legion of Racers (CT Hub) + SRA-SG; grassroots, no federation sanctioning.

Malaysia Toyota GR eRacing, jointly sanctioned by the national motorsport & esports federations.

Indonesia region's standout individual talent (Maulana); community-led, not federation-run.

Australia Motorsport Australia runs its own esports hub — the region's deepest federation integration.

THE NOTE THIS BOOK ENDS ON

Two national esports titles from a studio on the sixth floor of a Kallang Avenue office block is not a small thing for a hobby this specialised, in a country this size. It is also, honestly, most of what can currently be verified — which is itself a fair place for an ultra deep dive to leave a live, ongoing story.⁴

SOURCES 1 simracing.sg/home; simracing.sg/membership-join; simracing.sg/tag/iracing. 2 velocity.toyota.com.my; thesimgrid.com/communities/msrl. 3 esports.motorsport.org.au/communities; esportsinsider.com, "Cash Converters Supercars Eseries." 4 Editorial synthesis, this edition, per the sourcing standard stated throughout. 5 [Liquipedia](https://liquipedia.net) Sim Racing Wiki, "Gran Turismo World Series/2026/Esports Team Competition."

THE LAST WORD

Every hobby this magazine has covered as a special edition shares one trait: the entry price keeps falling while the ceiling keeps rising, and the honest story sits in the gap between them, not at either end. Sim racing's gap is wider than most — a \$299 bundle and a \$50,000 motion rig run the identical software, on the identical Tuesday-night grid. That is either the most democratic thing about the hobby, or the thing that lets its most expensive corners pretend the cheap end doesn't exist. This book tried to hold both readings at once, honestly, all the way through.

FULL SOURCE LIST

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